



## MANIPAL ACADEMY OF HIGHER EDUCATION

### OPERATING SYSTEMS [CSE 3123]

**Marks: 30**

**Duration: 90 mins.**

#### MCQ

**Answer all the questions.**

Section Duration: 20 mins

- 1) Identify the components of a program from the following list which are shared across threads in a multithreaded process.  
  
i) Register values  
ii) Heap memory  
iii) Global variable  
iv) Stack Memory  
  
[i, ii](#) [i, ii, iii](#) [ii, iii](#) [i, ii, iii, iv](#)  
  
(0.5)
- 2) Which multithreading model allows true concurrency on multiprocessor systems?. BL2, Co1  
  
[Many-to-One](#) [One-to-One](#) [One-to-Many](#) [Cooperative model](#)  
  
(0.5)
- 3) Identify which scheduling algorithm suffers from the “convoy effect”  
  
[Round Robin](#) [FCFS](#) [Priority Scheduling](#) [Shortest Job First](#)  
  
(0.5)
- 4) There are 10 different processes running on a workstation. Idle processes are waiting for an input event in the input queue. Busy processes are scheduled with the Round-Robin time sharing method. Choose out of the following quantum times that is the best value for small response times, if the processes have a short runtime, e.g. less than 10ms?  
  
[tQ = 15ms](#) [tQ = 40ms](#) [tQ = 45ms](#) [tQ = 50ms](#)  
  
(0.5)
- 5) Select a solution to the problem of indefinite blocking of low-priority  
  
(0.5)

processe.

[Starvation](#) [Aging](#) [Convoy Effect](#) [Spinlock](#)

6) Deadlock can be modeled using

[State transition diagram](#) [Resource Allocation Graph](#) [Resource Request Graph](#) [Wait-for-graph](#)

(0.5)

7) In which binding scheme can a process be moved from one segment to another during execution?

[Compile-time binding](#) [Load-time binding](#) [Execution-time binding](#) [Static binding](#)

(0.5)

8) Swapping is primarily used to.

[Increase CPU speed](#) [Increase multiprogramming degree](#) [Reduce external fragmentation](#) [Handle deadlocks](#)

(0.5)

9) Which of the following allocation strategies can cause external fragmentation?.

[Paging](#) [Segmentation](#) [Continuous allocation](#) [None of the options](#)

(0.5)

10) Which of the following is NOT typically supported in mobile systems?

[Paging](#) [Swapping](#) [Dynamic loading](#) [Fragmentation](#)

(0.5)

## DESCRIPTIVE

**Answer all the questions.**

- 11) With a neat diagram, elaborate on how the operating system distinguishes between the execution of operating system code and user-defined code. (3)
- 12) Differentiate between direct communication and indirect communication. Mention all the properties of communication link in both type of communication. (3)
- 13) Illustrate and explain with a diagram how the CPU switches from a process to a process. Compare and contrast monolithic and microkernel architectures. (3)
- 14) Apply semaphore operations (wait and signal) to show how they can be used to control access to a printer shared by multiple processes. (2)

15)

Assume that the following processes are scheduled using a preemptive priority scheduling algorithm. Each process is assigned a numerical priority, with a smaller number indicating a higher relative priority

| Process | Arrival time | Burst Time | Priority |
|---------|--------------|------------|----------|
| P1      | 0            | 8          | 2        |
| P2      | 1            | 4          | 1        |
| P3      | 2            | 9          | 3        |
| P4      | 3            | 5          | 2        |

(4)

1. Construct the Gantt chart showing how the CPU is allocated over time.
2. Calculate the Turnaround Time , and Waiting Times for each process.
3. Find the Average Waiting Time (AWT) and Average Turnaround Time (ATAT).

16)

A computer system has 5 processes (P0–P4) competing for 3 resource types (A, B, C). Apply Banker's algorithm to do compute the following tasks.

| Processes | Max   | Allocation | Available |
|-----------|-------|------------|-----------|
|           | A B C | A B C      | A B C     |
| P1        | 1 0 2 | 3 2 2      |           |
| P2        | 2 1 0 | 4 2 2      |           |
| P3        | 3 1 2 | 6 3 4      | 2 1 2     |
| P4        | 0 2 1 | 2 3 3      |           |
| P5        | 1 1 1 | 4 2 3      |           |

(4)

- (a) Construct the Need Matrix for all processes.
- (b) Apply the Safety Algorithm to check if the system is currently in a safe state. Find the safe sequence.
- (c) If Process P1 makes a request (1,0,1), check whether the request can be granted immediately.

17)

Apply the different dynamic storage allocation strategies (first-fit, best-fit, worst-fit) to allocate memory holes of sizes 200 KB, 300 KB, and 100 KB to four processes arriving with memory requirements of P1 = 180 KB, P2 = 260 KB, P3 = 100 KB, P4 = 210 KB. Show which strategy is most efficient.

(3)

18)

Apply the idea of fragmentation to explain how compaction can reduce external (3)

fragmentation, and why it cannot eliminate internal fragmentation.



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